

CMVS – lab2

Controllability

- The reachability Gramian matrix is defined as follows:

$$G(0, T) = \int_0^T e^{A(T-\tau)} B B^T e^{A^T(T-\tau)} d\tau$$

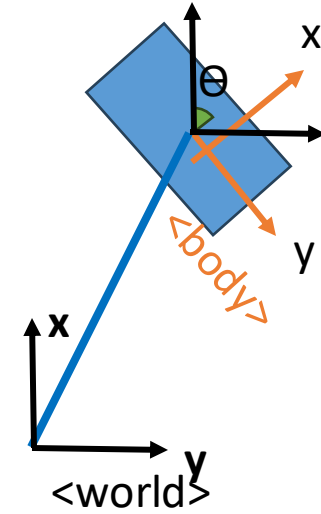
- Reachability Gramian theorem: A state is reachable in time T iff it belongs to the range of G(0,T). The linear system is completely reachable in time T iff G(0,T) has full rank.
- Continuous LTI systems: reachability $\Leftrightarrow$ controllability
- Computing the correspondent input:

$$u(t) = B^T e^{A^T(T-t)} G(0, T)^{-1} x_f$$

ROV with 3 dofs

$$\begin{aligned} m\ddot{x} &= -\alpha\dot{x} + F_X \\ m\ddot{y} &= -\beta\dot{y} + F_Y \\ J\dot{w} &= -\gamma w + N_Z \end{aligned}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$



$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$

ROV with 3 dofs



An ROV like the BlueROV2 is typically actuated by a number r of fixed thrusters.

In order to link the vector τ to the voltages V_i applied to each thruster i ($i \in \{1, \dots, r\}$), we use the expression:

$$\begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix} = \tau = T_{thruster} K(s) F(V)$$

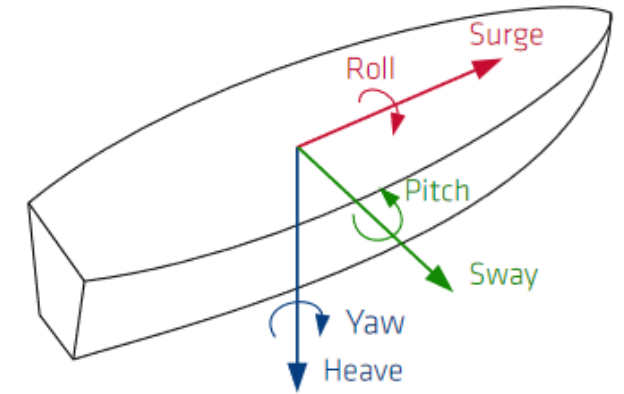
$V = [V_1, \dots, V_r]^T$ is the thrusters voltages

$F = [F_1, \dots, F_r]^T$ is the generated forces by the thrusters

$K(s)$ is the DC-gain transfer function for the dynamic relationship between V_i and F_i

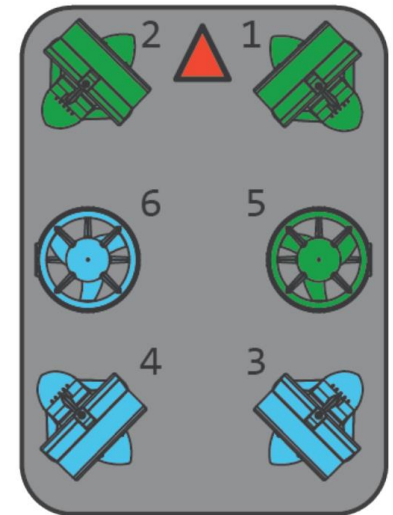
$T_{thruster}$ is the thruster allocation matrix

Thruster allocation matrix (TAM)



Thruster allocation matrix (TAM)

	T1	T2	T3	T4	T5	T6
Surge	0.7071	0.7071	-0.7071	-0.7071	0	0
Sway	-0.7071	0.7071	-0.7071	0.7071	0	0
Heave	0	0	0	0	-1	1
Roll	0	0	0	0	0.218	0.218
Pitch	0	0	0	0	0.12	-0.12
Yaw	-0.1888	0.1888	0.1888	-0.1888	0	0



x
y

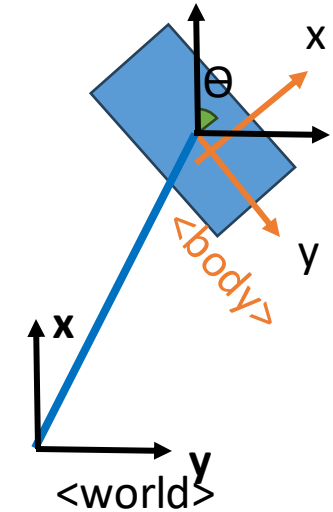
theta

ROV with 3 dofs

$$\tau = \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix} = T_{thruster}KV \approx \begin{bmatrix} 0.7071 & 0.7071 & -0.7071 & -0.7071 \\ -0.7071 & 0.7071 & -0.7071 & 0.7071 \\ -0.1888 & 0.1888 & 0.1888 & -0.1888 \end{bmatrix} \begin{bmatrix} 30 & 0 & 0 & 0 \\ 0 & 30 & 0 & 0 \\ 0 & 0 & 30 & 0 \\ 0 & 0 & 0 & 30 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

$$y = Mx$$

$$\tau = T_{thruster}K.V$$



We can reformulate the state equations by using as inputs the thruster voltages instead of the forces. To do this, consider the given state equations:

$$\frac{d}{dt}\mathbf{x} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{pmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ \omega \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{pmatrix} F_x \\ F_y \\ N_z \end{pmatrix}$$

Write $[F_x; F_y; F_z]$ as functions of the voltages:

$$\begin{pmatrix} F_x \\ F_y \\ N_z \end{pmatrix} = 30 \begin{bmatrix} 0.7071 & 0.7071 & -0.7071 & -0.7071 \\ -0.7071 & 0.7071 & -0.7071 & 0.7071 \\ -0.1888 & 0.1888 & 0.1888 & -0.1888 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

Putting everything together, we get

$$\frac{d}{dt} \mathbf{x} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ \omega \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 30/m & 0 & 0 \\ 0 & 30/m & 0 \\ 0 & 0 & 30/J \end{pmatrix} \begin{pmatrix} 0.7071 & 0.7071 & -0.7071 & -0.7071 \\ -0.7071 & 0.7071 & -0.7071 & 0.7071 \\ -0.1888 & 0.1888 & 0.1888 & -0.1888 \end{pmatrix}}_{\mathbf{B}} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{pmatrix}$$

Controllability

Exercise 1 (Lab2_ex1.m)

- 1.1. Define the system on MATLAB
- 1.2. Check if the system is controllable or not
- 1.3. If it is, compute the input such that initial state is $x_0 = [0;0;0;0;0;0]$ and final state is $x_f = [-1;2;3;0;0;0]$, in time $T = 20$ s
- 1.4. Simulate the system with the given input
- 1.5. Plot the state evolution

Controllability

1.6. Assume that thruster 2 is not working, that is, the second column of the TUM becomes a vector of zeros. How is controllability affected? Briefly comment.

1.7. Repeat by disabling thrusters 2 and 3.

1.8. In the case of not fully controllable systems, print matrices U , V and briefly comment about the controllability space.